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Endpoint-Calibrated Spatial Graph Rollouts for Probabilistic Functional Tumor Volume Forecasting in Breast DCE-MRI

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ABSTRACT Longitudinal breast dynamic contrast-enhanced magnetic resonance imaging (DCE-MRI) provides repeated tumor-burden measurements during neoadjuvant treatment, but most response models collapse this sequence into endpoint labels instead of forecasting future tumor states with calibrated uncertainty. This paper presents a synthetic-to-real evaluation of endpoint-calibrated spatial graph rollouts for functional tumor volume (FTV) forecasting. Tumors are represented as registration-consistent supervoxel graphs; a scheduled-sampling message-passing forecaster predicts spatial and imaging state evolution; endpoint calibration centers the rollout; and empirical residual Monte Carlo simulation converts the graph center into patient-level FTV distributions. Controlled synthetic tumors first test when graph neighborhoods should help. In the latent spatial-field setting, where neighborhoods reveal hidden response structure, the final hybrid-edge model reduced $T_0 \rightarrow T_3$ MC-mean FTV error by 5.43 mL and CRPS by 3.53 relative to a no-edge control. The same framework was then evaluated on 758 held-out patients from the I-SPY2/ACRIN graph-bearing cohort. Relative to the original graph rollout, the final model reduced deterministic FTV error from 12.88 mL to 4.63 mL, MC-mean FTV error from 15.26 mL to 5.56 mL, and conformal 90% interval width from 59.51 mL to 22.38 mL, while improving raw 90% coverage from 0.766 to 0.877. Its Monte Carlo samples also identified low residual MRI burden at T_3 (FTV < 5 mL) with AUC 0.957, comparable to strong last-observed FTV threshold baselines. These results support calibrated structured tumor-state forecasting: message passing is most useful when graph neighborhoods carry response structure, endpoint calibration makes spatial rollouts clinically interpretable, and scalar or hybrid readouts remain appropriate when the clinical query is only future FTV.

INDEX TERMS Breast DCE-MRI, conformal prediction, functional tumor volume, graph neural networks, longitudinal imaging, Monte Carlo simulation, uncertainty calibration.

I. INTRODUCTION

SERIAL dynamic contrast-enhanced magnetic resonance imaging (DCE-MRI) during neoadjuvant breast cancer treatment creates a forecasting problem: given the tumor observed so far, estimate future active burden and the uncertainty around that burden. The I-SPY program established serial breast MRI as a response-monitoring substrate for adaptive neoadjuvant trials [1]. Within that setting, functional tumor volume (FTV) is a central quantitative MRI biomarker, and longitudinal FTV and related DCE-MRI features carry response and recurrence information [2]–[5]. Public I-SPY2 and ACRIN-6698 imaging resources make this problem testable at cohort scale [6], [7].

The clinical value of this forecasting task goes beyond reducing numerical FTV error. During neoadjuvant therapy, serial MRI is used to monitor whether tumor burden is falling, persisting, or remaining uncertain. A calibrated future FTV distribution can express both the expected residual burden and the uncertainty around that burden, which is useful for trial analytics, response monitoring, and identifying cases whose imaging trajectory remains high-risk or poorly resolved. This paper therefore treats FTV forecasting as an uncertainty-aware imaging-burden problem rather than as only a retrospective response classification task.

The spatial structure of the tumor is central to that problem. A breast tumor seen on DCE-MRI is not a single homogeneous object: enhancement, washout, necrosis, boundary motion, and segmentation uncertainty can vary across the three-dimensional lesion. Representing the tumor as a graph of local supervoxel regions gives each region its own state while preserving local neighborhood relationships. In that representation, forecasting asks how the local tumor regions move, change enhancement, and remain active or become inactive over future visits. The graph is therefore not used only as a machine-learning convenience; it is a compact way to represent a spatially organized tumor state that can be rolled forward and queried for future FTV.

Existing breast MRI prediction work covers several related but distinct directions. Multi-feature MRI models and longitudinal fusion networks predict pathologic complete response or related response endpoints [4], [8]. More recent spatiotemporal MRI classifiers explicitly model temporal or regional imaging interactions for early response prediction [9], [10]. Mechanistic breast cancer digital twins take a different route, calibrating patient-specific tumor growth or treatment-response equations from imaging and clinical observations [11]–[15]. Together, these studies show that serial MRI contains clinically meaningful response information, but most learning-based approaches reduce the longitudinal image sequence to scalar features, a future response label, or a single burden estimate.

That framing is useful for response classification, but it leaves out two properties needed by a forecasting system: a patient-specific spatial tumor state and a calibrated distribution over the future endpoint. This paper asks a spatial version of the problem. Each tumor is represented as a registration-consistent graph of three-dimensional supervoxels. The model rolls this graph forward by predicting regional motion, imaging-feature change, and active or inactive status, then queries the forecasted graph for future FTV and sampled tumor-cloud uncertainty. The graph rollout builds on message-passing and learned physical-simulation ideas, where local interactions are used to evolve structured states over space or time [16], [17]. The uncertainty layer is evaluated with coverage, conformal intervals, and CRPS because forecast distributions should be judged by both calibration and sharpness [18]–[21].

Graph message passing is a plausible inductive bias because tumor response is spatially organized. Neighboring regions may share enhancement behavior, perfusion, necrosis, boundary shrinkage, registration uncertainty, or local segmentation noise. Its value, however, depends on whether the chosen edges carry response information beyond each node's own features. If isolated node features already contain the relevant future-response signal, or if the edge set mixes unrelated tumor regions, graph neighborhoods may add little or even inject noise. The first test in this paper is therefore synthetic: controlled graph tumors are generated under node-local, neighbor-coupled, and latent spatial-field dynamics so that the condition under which message passing should help is known.

The real-data model then applies the same idea to the I-SPY2/ACRIN graph-bearing cohort. The original scheduled-sampling graph rollout produced coherent three-dimensional tumor clouds, but its deterministic center was too low in FTV and active tumor mass. This exposed the central separation studied here: geometric plausibility and endpoint calibration are separable properties. The first corrected model, Endpoint+Active, adds FTV and active-burden supervision to test whether a spatial rollout can become a calibrated center for residual Monte Carlo (MC) uncertainty. The final retained model, Hybrid-Edge $k = 8$, then updates the neighborhood and edge-attribute design after testing whether message passing becomes more useful when graph edges encode local spatial proximity and imaging-feature similarity.

The paper is organized around one synthetic-to-real claim: graph neighborhoods are valuable when they carry local response information, but the resulting spatial rollout must still be calibrated to the clinical endpoint. The contribution can be read as three linked claims:

- *Graph-native tumor-state forecasting:* the predicted object is a future tumor graph state with active-node, imaging-feature, and sampled-cloud structure, not only a scalar burden estimate.
- *Endpoint calibration:* FTV and active-burden supervision convert the rollout from a plausible geometric forecast into a calibrated FTV center.
- *Residual uncertainty evaluation:* empirical MC and conformal readouts quantify coverage, sharpness, subgroup calibration, and clinically interpretable MRI-burden probabilities.

The central claim is therefore layered: structured tumor-state forecasting, endpoint calibration, and scalar or hybrid endpoint readouts are distinct components of longitudinal imaging forecasts and should be evaluated together. This framing is intended to make the paper readable as a forecasting study rather than as a pure architecture paper: each experiment asks whether one of these layers is necessary, calibrated, or clinically interpretable.

II. RELATED WORK AND CONTRIBUTION

A. LONGITUDINAL BREAST MRI AND FTV

Serial breast MRI response modeling has a strong clinical foundation. I-SPY2 was designed as an adaptive neoadjuvant platform trial with serial imaging and biomarker collection [1]. Earlier I-SPY analyses established DCE-MRI and FTV as informative longitudinal measures of tumor response and longer-term risk [2], [3]. Later I-SPY2 analyses showed that combining multiple MRI features, including FTV, improves response prediction compared with using individual features alone [4]. Protocol-adherence analyses further showed that FTV performance depends on standardized acquisition and image quality, and more recent work has emphasized that longitudinal FTV estimation error itself matters when MRI is used for response monitoring and trial assessment [5], [22]. These studies motivate FTV as the endpoint that the rollout distribution must be calibrated to, rather than treating the supervoxel forecast as only a geometric prediction problem.

B. DEEP LEARNING FOR BREAST MRI RESPONSE

Most published breast MRI response models operate at the patient or image level: handcrafted MRI features, radiomics, convolutional models, transformers, or multimodal predictors are used to estimate a response endpoint from one or more visits. Recent longitudinal and spatiotemporal MRI deep-learning work demonstrates the value of temporal representation learning, but still primarily maps one or more scans to pathologic complete response (pCR) or another scalar response label [8]–[10]. That framing is clinically valuable, while leaving open the forecasting question studied here: whether a model can produce a patient-specific three-dimensional tumor trajectory and a calibrated distribution over future imaging burden.

Table 1 summarizes the task positioning relative to longitudinal FTV biomarker studies, breast MRI response classifiers, mechanistic digital-twin models, and the direct graph-rollout baseline.

This positioning also separates the present task from nearby IEEE imaging traditions. Recent IEEE work in breast DCE-MRI has advanced image-native tumor segmentation with hybrid convolutional/transformer architectures [23], while future-MRI prediction work has shown that temporal generative models can synthesize missing or future sessions in other longitudinal MRI settings [24]. Those directions are important but solve different problems: segmentation returns the present tumor support, and future-image synthesis returns an image-like target. The target here is a calibrated future FTV distribution together with a graph-native spatial tumor state. This makes the novelty narrower and more precise: probabilistic longitudinal FTV forecasting with a graph-native state and calibrated endpoint burden readout.

C. DIGITAL TWINS AND IMAGE-BASED TUMOR FORECASTING

Digital twin work is the closest conceptual neighbor. In oncology, a digital twin is generally a patient-specific model updated from individual data to simulate tumor behavior under possible interventions [25]. In breast cancer, image-based mathematical models have used longitudinal MRI to constrain reaction–diffusion and mechanically coupled tumor-growth equations for patient-specific response prediction [11], [12], [26]. More recent MRI-calibrated digital models forecast spatiotemporal triple-negative breast cancer response and evaluate alternative neoadjuvant schedules [13]–[15].

This paper is adjacent to that digital-twin literature but makes a different technical choice. Rather than calibrating a mechanistic PDE model of tumor cell growth and drug response, it learns a registration-consistent supervoxel graph rollout from the I-SPY2/ACRIN-derived cohort. Rather than optimizing treatment schedules, it focuses on imaging-burden forecasting across the observed MRI visits. Rather than using only a deterministic forecast, it builds empirical patient-level FTV distributions from held-out residual structure and evaluates their calibration. The result is a data-driven imaging forecast module that could later serve as one component of a longitudinal imaging digital-twin workflow after external and prospective validation.

D. GRAPH ROLLOUTS AND PROBABILISTIC CALIBRATION

Graph neural networks provide a natural inductive bias for relational and dynamical systems [16], [17], and scheduled sampling addresses exposure bias in autoregressive sequence prediction [27]. This work combines those ideas with registration-consistent tumor supervoxels and endpoint-specific calibration. The graph component is central to the architecture. A breast tumor is treated as a longitudinal supervoxel graph whose nodes preserve regional identity across visits, then a local spatial graph convolution forecasts node displacement, imaging-feature change, and active-node state. This differs from image-level CNN or transformer response models because the output remains a structured tumor state that can be rolled forward and interrogated geometrically.

The probabilistic component is distinct from typical classification uncertainty and should be interpreted as empirical residual Monte Carlo rather than Bayesian posterior sampling over graph-network weights. After training, model weights are fixed; uncertainty samples are generated by resampling held-out residual structure around the deterministic rollout center. This follows bootstrap prediction-interval logic [28], [29] and related neural-network regression interval work that calibrates deterministic or probabilistic predictors using validation residuals or conformal prediction [19], [30]. The resulting FTV distributions are evaluated with coverage, conformal correction [20], [21], and Continuous Ranked Probability Score (CRPS), a proper scoring rule for full predictive distributions [18].

The contribution is not any single module alone, but the combined forecasting system and evaluation: a registration-consistent graph rollout architecture calibrated to endpoint-level FTV and active burden, with patient-level conditional Monte Carlo uncertainty for future MRI tumor burden.

III. PROBLEM FORMULATION AND FORECASTING FRAMEWORK

The main idea is simple: a tumor cloud can look spatially reasonable while its total active burden is still wrong. We therefore evaluate a rollout at two levels: the spatial tumor state and the scalar endpoint read from that state. Let $\widehat{G}_{p,t}$ denote the predicted tumor graph state and $\widehat{e}_{p,t} = R(\widehat{G}_{p,t})$ denote a scalar endpoint readout, here FTV computed from predicted node volumes and active-node probabilities. A graph rollout can be spatially plausible if $d_G(\widehat{G}_{p,t}, G_{p,t})$ is small for a cloud metric such as SWD or Chamfer distance, while still being clinically miscentered if $\widehat{e}_{p,t}$ is biased. For patient-level uncertainty, the requirement is stronger: the predictive distribution $\widehat{P}(e_{p,t} \mid G_{p,\leq s})$ should be sharp, centered, and well-covered for the endpoint being queried.

This gives the paper three recurring questions. Does the rollout preserve tumor-cloud geometry? Is the endpoint readout calibrated to observed FTV and active burden? Does the residual uncertainty layer improve coverage and CRPS without

TABLE 1. Compact task positioning relative to longitudinal breast MRI response models and mechanistic digital-twin studies. Recent breast MRI deep-learning models mainly predict pCR or response labels; this paper instead forecasts future FTV distributions and tumor graph states.

Comparison family	Representative work	Main output	Relevance and gap
FTV biomarker studies	I-SPY and I-SPY2 MRI/FTV analyses [2]–[4]	pCR, recurrence-risk, or response association	Establishes FTV as clinically meaningful; the present work adds patient-level forecast distributions for future FTV.
Longitudinal MRI fusion models	Huang et al. longitudinal breast MRI response modeling [8]	Patient-level pCR or response label	Uses serial MRI to improve response classification; the present work extends the target to tumor graph states and calibrated future FTV distributions.
Spatiotemporal MRI classifiers	Tang et al. spatiotemporal interaction modeling and BSTNet-type work [9], [10]	Patient-level pCR or response probability	Captures temporal/spatial MRI patterns for classification, but the prediction target remains a response label rather than a future tumor-burden trajectory.
Mechanistic breast digital twins	MRI-calibrated reaction–diffusion and response models [11]–[15]	Patient-specific response simulation or regimen optimization	Shares the patient-updated forecasting goal, but uses mechanistic equations and treatment-response simulation rather than learned registration-consistent graph rollouts with empirical FTV uncertainty.
Current graph rollout baseline	Scheduled-sampling graph-convolutional rollout	Future spatial tumor cloud and deterministic FTV center	This is the direct comparator; endpoint FTV/active-burden supervision corrects the biased tumor-burden center before the residual MC sampler is applied.

making intervals unnecessarily wide? The ablations answer these questions in order, moving from a geometry-only rollout to endpoint-calibrated, edge-aware, scalar, and hybrid readouts. Figure 1 summarizes this layered evaluation stack.

At the patient level, the forecasted object is a future three-dimensional tumor state, not a response label. Each node represents a transported baseline tumor region; node features describe local volume, enhancement, and activity; and edges connect nearby regions or regions with similar imaging features. The model first predicts the next graph state. FTV is then read from that predicted graph by summing the expected active-volume contribution of the nodes. This separation is useful for evaluation: spatial graph quality, endpoint FTV calibration, and residual predictive uncertainty can be diagnosed independently.

IV. SYNTHETIC VALIDATION FRAMEWORK

The synthetic experiments were added to test the graph assumption before interpreting the real-cohort ablations. A message-passing model is expected to outperform a nodewise forecaster when the future state of one tumor region depends on local spatial context that is only partially visible from that node's own features. The synthetic generator therefore creates persistent supervoxel graphs across T_0, T_1, T_2, and T_3 with known FTV, active-node-count, and response-class trajectories, then changes only the data-generating dependence structure.

These simulations serve as mechanism checks under known dependence structure. They identify the conditions where a graph model improves over, matches, or falls behind a no-edge forecaster before the same design is tested on real longitudinal breast MRI graphs.

Figure 2 summarizes the three synthetic regimes used to evaluate this condition. The node-local cohort is a negative control: future node survival and volume change are mostly recoverable from each node's own features, so the no-edge model should be competitive. The neighbor-coupled cohort makes future survival and volume depend partly on local neighbor active-node state, resistance, and burden. The latent spatial-field cohort is the strongest test of message passing: a smooth hidden regional resistance field drives future response, while each node observes only a noisy local proxy. In that setting, neighborhood aggregation can denoise the latent field and improve both endpoint uncertainty and sampled spatial-state behavior.

All synthetic cohorts are evaluated with the same terminal $T_0 \rightarrow T_3$ question used in the real cohort. The primary comparison is between a no-edge endpoint model and Hybrid-Edge $k = 8$, with full-graph, radial, and hybrid-neighborhood variants retained as intermediate controls. The same metric families are used throughout: deterministic FTV MAE, MC-mean FTV MAE, CRPS, raw 90% interval coverage and width, active-node-count error, SWD, Chamfer distance, and Dice. This makes the synthetic section a mechanism test rather than a separate toy problem.

V. MATERIALS AND METHODS

A. REAL COHORT AND GRAPH REPRESENTATION

The analysis uses 758 patients with four longitudinal DCE-MRI visits and usable registration across T_0, T_1, T_2, and T_3. These patients come from the four-visit graph-bearing I-SPY2/ACRIN-derived cohort and are evaluated under the same five-fold held-out-patient protocol used for the scheduled-sampling baseline. Each patient is predicted by the checkpoint from the fold in which that patient was held out.

The graph construction keeps the representation interpretable. Each node is a small baseline tumor region that is followed through time, and each edge describes a local relationship that the message-passing model can use during rollout. Each baseline tumor is partitioned into compact three-dimensional Simple Linear Iterative Clustering (SLIC) supervoxels [31]. Deformable registration transports the baseline supervoxel partition to later visits using symmetric diffeomorphic transforms [32]. As a result, node i denotes the same baseline tumor region at all visits, while node features are measured from the native follow-up image support.

For patient p and visit t, the graph is $G_p^{(t)} = (V_p^{(t)}, E_p^{(t)})$. Each node has six imaging-burden features: $\log(1 + \mathrm{voxel\ count})$, volume in mL, percent enhancement mean and standard deviation, and signal enhancement ratio mean

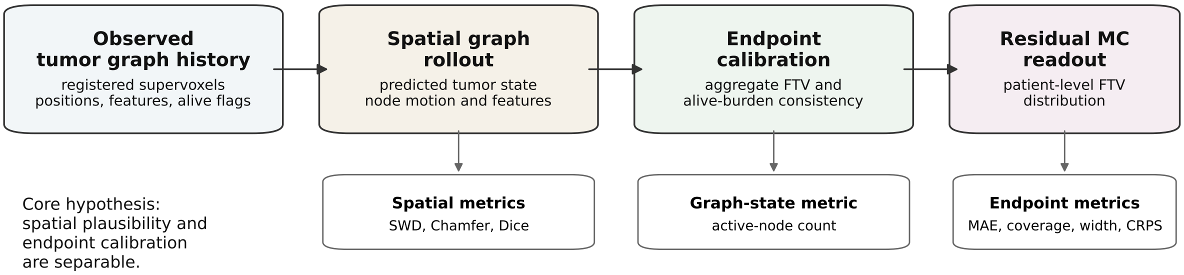

FIGURE 1. **Endpoint-calibrated spatial forecasting separates the spatial tumor state from the endpoint and uncertainty layers queried from that state. The graph rollout is evaluated as a sampled tumor cloud and active-node state, while the FTV readout is evaluated with endpoint error, coverage, sharpness, and CRPS.**

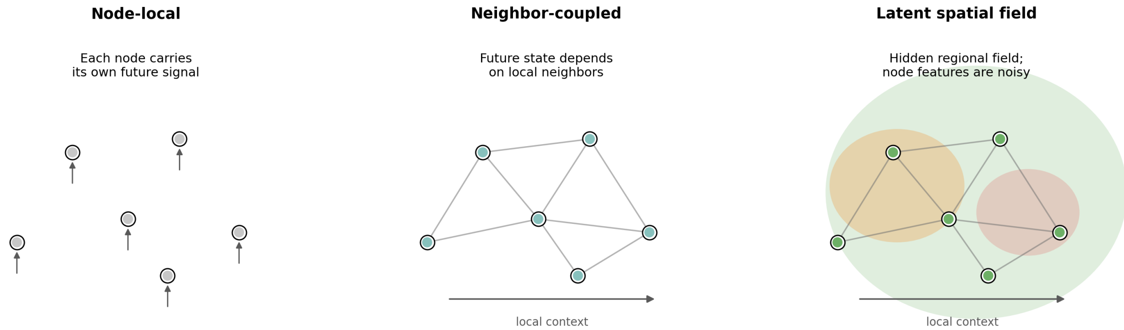

FIGURE 2. **Controlled synthetic tumor-graph regimes used before real-cohort evaluation. The node-local control tests whether message passing is unnecessary when isolated node features contain the relevant future-response information. The neighbor-coupled and latent spatial-field regimes introduce local dependence so that graph neighborhoods can carry predictive signal beyond node-local features.**

and standard deviation. Centroid positions are represented in millimeters after subtracting the visit-level tumor centroid. The active-node target records whether a transported baseline supervoxel still has tumor support at that visit.

Edges are rebuilt dynamically from the observed or self-fed rollout history. The first endpoint-calibrated model uses spatial $k = 8$ nearest-neighbor connectivity. The final retained Hybrid-Edge $k = 8$ model uses a more selective hybrid $k = 8$ neighborhood. Within each visit, candidate neighbors are ranked by an equally weighted normalized blend of spatial distance and feature distance, then symmetrized. Identity temporal edges connect the same transported supervoxel across observed history visits. Each retained edge also receives radial imaging-feature attributes describing distance, radial shell similarity, visit gap, and local imaging-feature contrast. The forecaster then predicts next-visit node displacement, feature change, and a pre-sigmoid active-node score. SLIC was used because it gives compact local regions with interpretable volume and enhancement summaries, and $k = 8$ was chosen to preserve local tumor-cloud connectivity without making the graph nearly dense. Centering positions at the visit-level tumor centroid makes the displacement target focus on within-tumor geometry rather than scanner-frame translation. For notation, p indexes patients, t indexes visits, and i indexes supervoxels. $V_p^{(t)}$ denotes the transported supervoxel nodes for patient p at visit t, $E_p^{(t)}$ denotes the corresponding dynamic edges, hats denote model predictions, and $\mathrm{FTV}_{p,t}$ denotes the observed functional tumor volume for patient p at visit t. Figure 3 summarizes the retained graph-forecasting and uncertainty workflow.

Figure 4 gives two illustrative successful held-out $T_0 \rightarrow T_3$ cases from the final model: one I-SPY2 case and one ACRIN case with non-trivial follow-up burden, raw interval coverage, and strong deterministic agreement. These examples show how the graph rollout is read at the patient level; the cohort-level claims are made from the aggregate tables and calibration figures rather than from these two cases alone.

B. ENDPOINT-CALIBRATED ROLLOUT TRAINING

The training changes are easiest to read in two parts. The first part defines how the graph passes messages between tumor regions. The second part defines how the rollout is centered on the clinical endpoint, FTV, while still preserving the spatial cloud losses from the original model.

All retained graph-family models are based on the scheduled-sampling consistent graph forecaster, which is the

Stage	Function in the retained pipeline
Registered graphs	Registration-consistent supervoxel states with shared node identities.
Hybrid-edge LSGC rollout	One graph forecaster reused for each autoregressive state transition, with dynamic hybrid spatial-feature neighborhoods in the final retained model.
FTV/active calibration	Aggregate burden losses center the deterministic FTV and active-node rollout.
Residual Monte Carlo	The unchanged residual sampler converts the calibrated center into patient-level FTV distributions.

FIGURE 3. Forecasting and uncertainty workflow. Registration-consistent supervoxel graphs are rolled forward with a shared LSGC forecaster. The final retained model uses dynamic hybrid spatial-feature neighborhoods with radial imaging-feature edge attributes. Aggregate FTV and active-burden objectives calibrate the deterministic rollout center before the unchanged residual Monte Carlo sampler generates patient-level FTV distributions.

first graph-convolutional rollout baseline developed for this task. The architecture embeds the six node features into a 64-dimensional hidden state, applies two residual local spatial graph convolution (LSGC) layers, and concatenates an 8-dimensional visit-gap embedding before three per-node heads: 3D displacement, six-dimensional feature change, and a pre-sigmoid active-node score. The LSGC layer is a continuous-filter message-passing operator over the patient tumor graph. For an edge $j \to i$, the filter is parameterized by the millimeter displacement between supervoxel centroids, the visit-time difference, the unit spatial direction, and optional retained edge attributes. In the final Hybrid-Edge $k = 8$ model, within-visit neighbors are chosen by

$$s_{ij} = \alpha \frac{\|c_i - c_j\|_2}{m_c} + (1-\alpha)\frac{\|f_i - f_j\|_2}{m_f}, \tag{1}$$

where c_i is the centroid coordinate, f_i is the normalized six-feature node vector, and m_c and m_f are median positive distances used only for scale normalization. We fixed $\alpha = 0.50$ so that spatial proximity and imaging-feature similarity contributed equally after median-distance normalization. This choice is a neutral prior rather than a test-set-tuned weight: $\alpha = 1$ would reduce the rule to spatial nearest neighbors, while $\alpha = 0$ would ignore anatomy and rank only by feature similarity. The equal-weight setting therefore tests whether a simple local spatial-plus-imaging-feature neighborhood is useful before introducing another tunable hyperparameter. The $k = 8$ lowest-score neighbors are retained for each node and then symmetrized. For those edges, the radial imaging-feature attribute vector b_{ij} contains normalized centroid distance, absolute and signed radial-shell difference, source and target radius, radial cosine similarity, a same-shell kernel, absolute visit gap, temporal-edge indicator, volume-feature contrast, percent-enhancement/SER contrast, and total feature contrast. The edge descriptor z_{ij} used by the LSGC filter is

$$z_{ij} = \big[\mathrm{RBF}(\|\Delta x_{ij}\|), \{\sin(\omega_k \Delta t_{ij}), \cos(\omega_k \Delta t_{ij})\}_{k=1}^{K}, \widehat{d}_{ij}, b_{ij}\big], \tag{2}$$

Paper term	Implementation setting	Role
Hybrid-Edge $k = 8$	`hybrid_a50_bio_k8`	Final retained graph model used for the main reported endpoint, uncertainty, and graph-state results.
Endpoint+Active	`spatial` $k = 8$ `+ FTV/active losses`	Endpoint-calibrated comparator that keeps the original spatial neighborhood.
Hybrid neighborhood	`hybrid_spatial _feature_a50`	Ranks candidate edges with a neutral equal-weight blend of normalized spatial and feature distances ($\alpha = 0.50$).
Graph degree	`k_spatial=8`	Retains the eight lowest-score within-visit neighbors per node before symmetrization.
Edge attributes	`radial_bio`	Appends radial, temporal, and imaging-feature contrast attributes to the LSGC filter input.
Residual MC	Same empirical residual sampler for all graph centers	Tests whether model changes improve the deterministic center under a fixed uncertainty layer.

TABLE 2. Implementation map for the retained graph-family models. The table connects manuscript terminology to the run settings used in the reported experiments while keeping the mathematical definition in the surrounding text.

where the sinusoidal terms encode the visit gap at K learned frequencies ω_k. This basis follows common continuous-coordinate neural modeling practice: radial basis expansions localize Euclidean distances for continuous-filter graph convolutions, and sine/cosine features encode ordered or continuous time differences across frequencies [33]–[35]. The message and node update are

$$\begin{aligned} m_{ij} &= \phi_\theta(z_{ij}) \odot W_m h_j, \\ h_i' &= W_s h_i + W_o \sum_{j \in \mathcal{N}(i)} m_{ij}. \end{aligned} \tag{3}$$

Here h_i is the hidden state for supervoxel i, Δx_{ij} is the relative centroid displacement, Δt_{ij} is the visit gap, $\widehat{d}_{ij}$ is the unit direction, ϕ_θ is the learned edge-filter network, RBF denotes radial basis function expansion, and $\odot$ denotes channel-wise multiplication. For earlier no-attribute models, b_{ij} is omitted from z_{ij}; the final Hybrid-Edge $k = 8$ model appends the radial imaging-feature attributes above. This lets the same local operator use spatial neighborhood structure, visit-gap information, and local imaging-feature contrasts when forecasting the next tumor-graph state. The graph-family models are initialized from the selected scheduled-sampling checkpoint, trained with AdamW at learning rate 10^{-4}, run with the backbone frozen for the first eight epochs, and trained for up to 120 epochs with early stopping.

Table 2 maps the paper-facing model terms to the implementation settings used for the retained run and the main comparator. The substring `bio` is an internal run tag for radial imaging-feature attributes and endpoint-burden losses; it is not used here to claim that the edge variables are biological markers.

The original one-step loss penalized node displacement, feature change, and cloud-level rollout geometry. The endpoint-

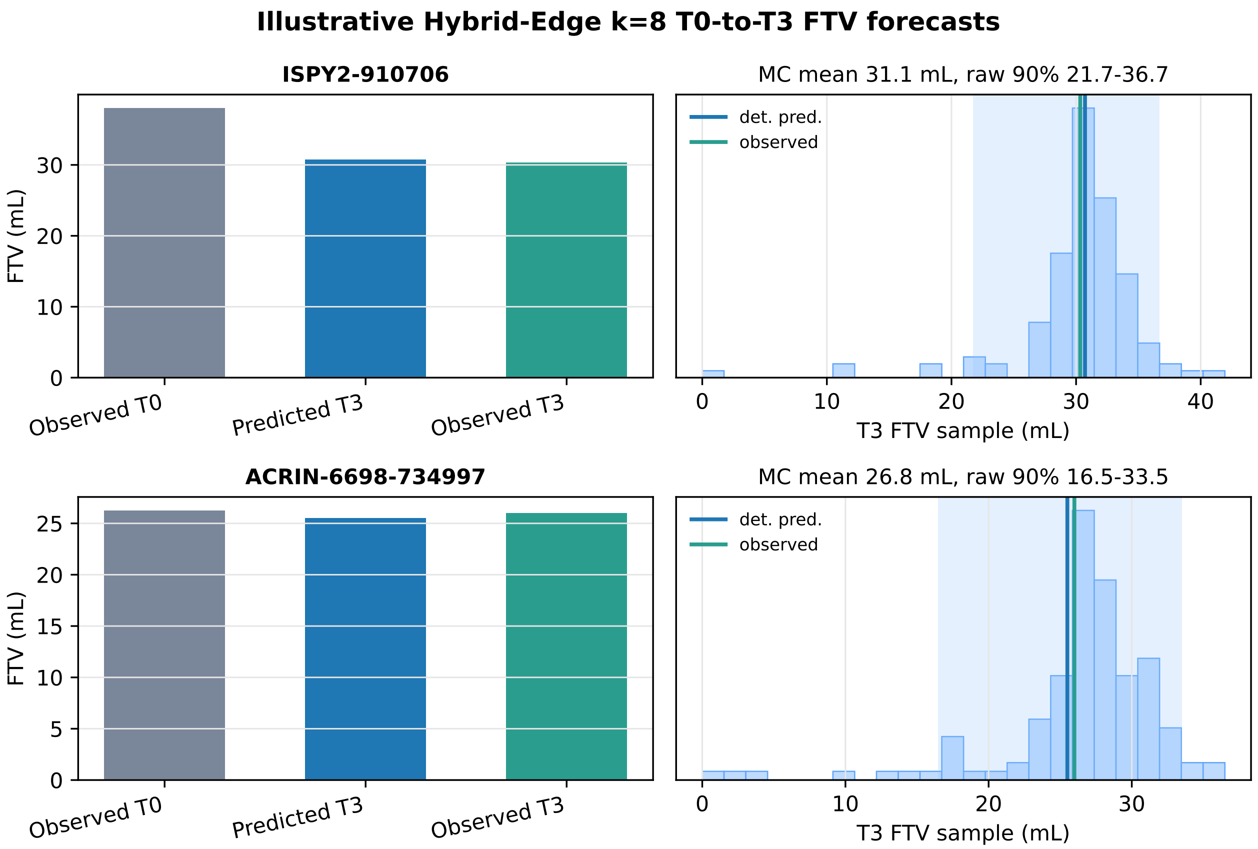

FIGURE 4. Illustrative successful held-out $T_0 \to T_3$ FTV forecast examples generated from the final retained Hybrid-Edge $k = 8$ outputs. Each row shows observed T_0 FTV, deterministic predicted T_3 FTV, observed T_3 FTV, and the model's Monte Carlo T_3 FTV distribution. The graph model predicted T_3 FTV of 30.72 mL versus 30.33 mL observed for ISPY2-910706 and 25.50 mL versus 25.99 mL observed for ACRIN-6698-734997.

calibrated model keeps those terms and adds two burden-level constraints: active-node mass and FTV. For a transition into target visit t, let $y_{i,t} \in \{0, 1\}$ denote whether transported supervoxel i is active at the target visit, and let $\mathcal{A}_{p,t} = \{i : y_{i,t} = 1\}$. Let $\Delta x_{i,t}$ and $\Delta f_{i,t}$ be the observed next-visit displacement and feature change, with hats denoting model predictions. The teacher-forced node losses are

$$\begin{aligned} \mathcal{L}_{\text{pos}} &= \frac{1}{|\mathcal{A}_{p,t}|} \sum_{i \in \mathcal{A}_{p,t}} \|\Delta \widehat{x}_{i,t} - \Delta x_{i,t}\|_2^2, \\ \mathcal{L}_{\text{feat}} &= \frac{1}{|\mathcal{A}_{p,t}|} \sum_{i \in \mathcal{A}_{p,t}} \|\Delta \widehat{f}_{i,t} - \Delta f_{i,t}\|_2^2. \end{aligned} \tag{4}$$

During self-fed scheduled-sampling steps, the cloud term compares the predicted and observed target point clouds:

$$\mathcal{L}_{\text{cloud}} = \mathrm{SWD}(\widehat{X}_{p,t}, X_{p,t}), \tag{5}$$

where $\widehat{X}_{p,t}$ and $X_{p,t}$ are the predicted and observed target-visit centroid clouds. SWD projects both clouds onto random unit directions and compares the resulting one-dimensional empirical quantiles:

$$\mathrm{SWD}(A, B) = \frac{1}{LQ} \sum_{\ell=1}^{L} \sum_{q=1}^{Q} \left| F^{-1}_{\theta_\ell^\top A}(\tau_q) - F^{-1}_{\theta_\ell^\top B}(\tau_q) \right|, \tag{6}$$

where θ_ℓ is projection direction ℓ, τ_q is quantile level q, and L and Q are the numbers of projections and quantiles. SWD was used because the predicted and observed tumor clouds are unordered and can contain different active-node sets; it compares their spatial distributions without requiring one-to-one node matching.

Predicted FTV is computed from the predicted node volumes and active-node probabilities:

$$\widehat{\mathrm{FTV}}_{p,t} = \sum_{i \in V_p^{(t)}} \widehat{v}_{i,t}\, \sigma(\widehat{a}_{i,t}), \tag{7}$$

where $\widehat{\mathrm{FTV}}_{p,t}$ is the model-predicted FTV, $\widehat{v}_{i,t}$ is the predicted volume contribution for supervoxel i in mL, $\widehat{a}_{i,t}$ is the raw active-node score, and $\sigma(\widehat{a}_{i,t})$ is the predicted active-node probability. The product is the expected active-volume contribution of node i. The endpoint losses are active-node binary cross-entropy (BCE), active-mass error, and FTV error:

$$\begin{aligned}\mathcal{L}_{\text{active BCE}} &= -\frac{1}{|V_p^{(t)}|}\sum_{i\in V_p^{(t)}}\Big[y_{i,t}\log\sigma(\widehat{a}_{i,t}) \\ &\qquad + (1-y_{i,t})\log(1-\sigma(\widehat{a}_{i,t}))\Big], \\ \mathcal{L}_{\text{active mass}} &= \left|\sum_{i\in V_p^{(t)}}\sigma(\widehat{a}_{i,t}) - \sum_{i\in V_p^{(t)}} y_{i,t}\right|, \\ \mathcal{L}_{\text{FTV}} &= \left|\log(1+\widehat{\text{FTV}}_{p,t}) - \log(1+\text{FTV}_{p,t})\right|. \end{aligned} \tag{8}$$

Here $\text{FTV}_{p,t}$ is the observed FTV. The $\log(1+x)$ transform keeps the FTV loss well-defined at zero and makes small and large tumors contribute meaningfully. The full loss, averaged over scheduled-sampling rollout steps, is

$$\begin{aligned}\mathcal{L} &= \mathcal{L}_{\text{pos}} + 0.5\mathcal{L}_{\text{feat}} + \mathcal{L}_{\text{cloud}} \\ &\quad + 0.05\mathcal{L}_{\text{active BCE}} + 0.05\mathcal{L}_{\text{active mass}} + 0.20\mathcal{L}_{\text{FTV}}.\end{aligned} \tag{9}$$

The numeric coefficients are the retained loss weights from the FTV/active-burden calibration grid rather than values tuned on the final test set. The feature-change coefficient was inherited from the scheduled-sampling baseline to keep feature regression secondary to coordinate and cloud consistency. The FTV term receives the largest endpoint weight because the diagnosed baseline failure was FTV under-centering; the active-node BCE and active-mass terms are smaller auxiliary constraints that stabilize node-level and aggregate activity without dominating the spatial rollout losses. The same retained weights are then held fixed for the edge-ablation family so that neighborhood changes are kept separate from loss retuning.

The schedules serve the same goal: keep the rollout comparable to the original baseline while gradually exposing the model to its own predictions. Scheduled sampling is enabled with maximum self-feeding probability 0.5 after a 10-epoch warmup and 60-epoch annealing schedule. The FTV/active-burden losses are annealed over 20 epochs. These schedule values were inherited from the scheduled-sampling baseline to preserve comparability and reduce early self-feeding instability while the endpoint losses ramp in. Checkpoint selection is still geometry-first, but adds small validation penalties for T_3 FTV error and active-node-count error. The first endpoint-calibrated configuration, Endpoint+Active, uses the strongest FTV/active-burden setting from the endpoint-calibration grid.

The final retained model, Hybrid-Edge $k = 8$, keeps these endpoint objectives and changes the graph-neighborhood construction. The original full graph allowed many weak edges, while no-edge and spatial-only ablations tested whether the node features alone could carry most of the scalar endpoint signal. The final model uses a hybrid $k = 8$ neighborhood: candidate neighbors are ranked by an equal-weight normalized blend of spatial proximity and feature similarity, and radial imaging-feature edge attributes encode distance plus local imaging-feature differences. The equal weighting is a neutral prior before validation, while $k = 8$ preserves local tumor-cloud connectivity without making the graph nearly dense. This choice was retained because it produced the best balance of endpoint accuracy, uncertainty sharpness, and preserved sampled-cloud geometry across the edge-ablation family, rather than because it minimized only one scalar metric.

C. CONDITIONAL MONTE CARLO SIMULATION

The MC layer is a fixed uncertainty wrapper around each deterministic rollout center. This design makes the comparisons interpretable: when MC performance improves, the change comes from a better graph center rather than from changing the residual sampler.

For each model, each held-out patient is evaluated under three conditioning regimes: T_0 observed; T_0,T_1 observed; and T_0,T_1,T_2 observed. This yields six prediction buckets: $T_0{\to}T_1$, $T_0{\to}T_2$, $T_0{\to}T_3$, $T_1{\to}T_2$, $T_1{\to}T_3$, and $T_2{\to}T_3$.

The deterministic rollout is first computed with the held-out fold checkpoint. Residuals between predicted and observed outcomes are pooled within each bucket, excluding the target patient. Each Monte Carlo draw resamples patient-level FTV, active-node count, global cloud displacement, local node displacement, and log-volume residuals, with Gumbel top-k active-mask stochasticity. Model weights, fold assignment, observed conditioning visits, and the deterministic center are fixed. Thus the raw MC intervals estimate uncertainty conditional on the trained rollout and empirical residual pool, while conformal expansion provides an additional distribution-free correction under the usual exchangeability assumption.

Each model generates

$$758 \times 6 \times 256 = 1{,}164{,}288$$

Monte Carlo samples, corresponding to 758 patients, six conditioning buckets, and $N_{\text{MC}} = 256$ draws per patient-bucket pair. Across the baseline and three biology-retrained models, the comparison covers 4,657,152 samples. The draw count was chosen as a practical compromise between stable patient-level quantiles and full-cohort computation across all folds and conditioning buckets. The subsequent edge-neighborhood and edge-attribute ablation used the same held-out residual MC protocol with $N_{\text{MC}} = 128$ for each model to support the larger ablation family under the available GPU budget. Raw 90% intervals use the 5th and 95th percentiles of sampled FTV. For the clinical forecasting summary, 90% was chosen as a conventional central prediction interval that is wide enough to assess uncertainty while remaining more interpretable than very high-coverage intervals. For conformal correction, let j index a held-out calibration patient in the same bucket, let $[\ell_j, u_j]$ be that patient's raw interval, and let y_j be the observed FTV. The nonconformity score is

$$s_j = \max\{\ell_j - y_j,\ y_j - u_j,\ 0\}. \tag{10}$$

The empirical 90% conformal quantile of $\{s_j\}$ within the same bucket is denoted $q_{0.90}$. For a target patient with raw interval $[\ell, u]$, the conformal interval is $[\ell - q_{0.90}, u + q_{0.90}]$. Sharpness and calibration are summarized by interval width, empirical coverage, and Continuous Ranked Probability Score (CRPS). Spatial samples are evaluated with the sliced Wasserstein distance (SWD) defined above [36], Chamfer distance, and Dice overlap. Dice is included as a conservative spatial-overlap diagnostic rather than as the primary endpoint. The forecasted object is a sparse transported-supervoxel state with predicted active nodes rather than a dense voxelwise segmentation mask. As a result, modest centroid or node-activation errors can produce low absolute Dice even when the sampled tumor cloud remains geometrically close under SWD or Chamfer distance. We therefore use Dice to check whether endpoint calibration destroys gross spatial overlap, while SWD and Chamfer are the main geometry metrics for unordered tumor-cloud forecasts. Paired uncertainty for reported model differences is estimated by 5,000 bootstrap resamples over patient identities; this count was used to stabilize paired confidence intervals while keeping the reporting pipeline reproducible.

Split and leakage controls were audited at the graph, residual, conformal, and scalar-baseline levels (Table 3). For graph training, fold k is held out at the patient level: the checkpoint for fold k is trained on patients with fold $\neq k$, validated on patients with fold $= k$, and normalized using statistics computed only from the training patients. The conditional MC evaluation then scores each patient with the checkpoint from the fold in which that patient was held out. Residual pools and conformal nonconformity scores are bucketed by conditioning and prediction visit, and the target patient is excluded before residual sampling or conformal expansion. For scalar and hybrid controls, residual means, residual MC pools, and hybrid ridge weights are fit on patients with fold $\neq k$ before being applied to fold k. The scalar features are limited to observed FTV history through the conditioning visit, and the hybrid center uses only the retained graph prediction, last-observed FTV, and T_0 FTV. Observed T_3 FTV is used as an outcome for scoring and as a training label for other folds; it is excluded from target-fold features.

D. REPRODUCIBILITY, DATA AVAILABILITY, AND GOVERNANCE

The reported experiments are designed to be auditable from the derived graph artifacts and held-out prediction tables. The public source imaging resources are the I-SPY2 and ACRIN-6698 breast DCE-MRI collections available through TCIA and their associated data-use terms. The reviewer-facing release repository associated with this manuscript is available at https://github.com/mirisr/breast-dce-mri-graph-rollouts. It includes preprocessing summaries, split files, training and evaluation scripts, retained-model configuration files, Monte Carlo evaluation code, and figure/table generation scripts. It also includes a supplementary reproducibility-details file that maps manuscript claims to model tags, split files, launchers, and table/figure artifacts. If journal policy requires an anonymous review package, an anonymized archive with the same aggregate tables and scripts can be provided during review. Protected patient-level imaging data and restricted derived files will not be redistributed through the code repository.

The split and leakage audit above defines the patient-level fold protocol, residual-pool exclusion rule, conformal calibration rule, and scalar/hybrid baseline fitting rule used for the main reported results. Retained-model hyperparameters are stated in the architecture and training descriptions, and Monte Carlo draw counts are reported separately for the final comparison ($N_{\mathrm{MC}} = 256$) and the broader edge-ablation screen ($N_{\mathrm{MC}} = 128$). Randomized components are controlled through fixed fold assignments, fixed model checkpoints for held-out evaluation, and fixed post-training residual MC protocols.

This work is a retrospective secondary analysis of de-identified longitudinal breast MRI resources. Source-study consent, trial governance, and data-access requirements are handled through the originating I-SPY2/ACRIN resources and TCIA access mechanisms. No prospective intervention, direct participant contact, treatment assignment, or identifiable-data analysis was performed as part of this study. No local protocol number is reported in this manuscript draft; any institution-specific exemption documentation can be supplied with the submission record if required by the journal or institution.

VI. EXPERIMENTS AND ABLATIONS

The experiments follow the same order as the paper's argument. First, synthetic cohorts test when message passing should help. Second, the real I-SPY2/ACRIN cohort tests whether endpoint and active-burden losses correct the original low-FTV bias without damaging spatial cloud metrics. Third, the empirical MC sampler is held fixed to test whether a better deterministic center improves coverage, interval width, and CRPS. Fourth, graph-native outputs are evaluated because active-node and sampled-cloud states are graph-specific outputs. Finally, edge-neighborhood ablations and stratified analyses test whether the final graph model is robust across edge definitions, cohorts, and molecular subtype groups. A small independent Breast-MRI-NACT-Pilot cohort is included only as an external stress test of paper-family model transfer; because it contains 11 graph-ready patients, it is not treated as powered clinical external validation. Expanded endpoint-loss, graph-edge, scalar-baseline, burden-conditional, reliability, source-stratified, and external stress-test tables are provided in the Supplementary Material so that the main paper can keep the narrative focused on the retained model and the key reviewer-facing comparisons.

The primary real-cohort analysis is held-out $T_0 \to T_3$ FTV forecasting on the graph-bearing I-SPY2/ACRIN cohort with Hybrid-Edge $k = 8$ as the retained graph model. The primary metrics are MC-mean FTV MAE, CRPS, raw 90% coverage, and conformal 90% interval width. Secondary analyses include deterministic FTV error and bias, active-node-count error, sampled-cloud SWD, Chamfer distance, Dice, $T_1 \to T_3$ and $T_2 \to T_3$ horizon checks, subtype/source/burden stratification, and low-residual MRI-burden threshold readouts. The pCR association is treated only as exploratory context because the model forecasts MRI burden rather than pathology response.

TABLE 3. Patient-level split and leakage audit for the reported FTV forecasting results. The audit covers graph training, residual MC, conformal correction, scalar baselines, and hybrid readouts used in the held-out analyses.

Audit item	Implementation check	Current-run evidence
Deterministic graph folds	Fold k checkpoints train on fold $\neq k$ patients, validate/score fold $= k$ patients, and compute normalization statistics on training patients only.	758 graph-ready held-out patients scored across five folds: 147, 154, 146, 151, and 160 patients.
Residual MC pools	Residuals are pooled within each start/predicted-visit bucket, and the target patient is removed before MC sampling.	Every retained-model bucket has 758 patients and 757 calibration patients per target.
Conformal calibration	Raw-interval nonconformity scores are computed within bucket; the target patient is excluded before computing the conformal quantile.	The retained run applies bucket-specific 90% conformal correction to all six conditioning buckets.
Scalar and hybrid baselines	Scalar residual means, scalar residual MC, and hybrid ridge centers are fit on fold $\neq k$ patients before scoring fold k.	Hybrid features are retained graph FTV, last-observed FTV, and T_0 FTV; no target T_3 feature is included.
Target-visit leakage	For start visit s, scalar trend features use only observed FTV visits with index $\leq s$; target FTV is used only as the scored outcome.	$T_0 \to T_3$ trend baselines reduce to carry-forward because only T_0 is observed.

VII. RESULTS

The results are reported from mechanism to application. Synthetic cohorts first show when graph neighborhoods contain useful information. The real-cohort sections then show how endpoint calibration, uncertainty calibration, and edge-aware message passing change the I-SPY2/ACRIN forecasts.

A. SYNTHETIC CONTROLS DEFINED WHEN MESSAGE PASSING HELPED

The synthetic experiments first tested whether graph neighborhoods were useful under known data-generating conditions. The node-local cohort behaved as expected: the no-edge model remained competitive for scalar FTV because each node's future response was mostly encoded in its own features. This negative control tests the boundary condition where message passing should add little.

The latent spatial-field cohort gave the strongest positive result. In that cohort, future survival and volume were driven by a hidden smooth regional response field, and each node saw only a noisy proxy. Under this condition, Hybrid-Edge $k = 8$ reduced $T_0 \to T_3$ MC-mean FTV MAE by 5.43 mL relative to the no-edge control, reduced CRPS by 3.53, narrowed the raw 90% interval by 24.14 mL, and improved MC Chamfer by 0.41 mm (Table 4). Figure 5 visualizes this synthetic mechanism: once the future state depends on a local spatial field, neighborhood-aware message passing improves the endpoint distribution and the sampled tumor-cloud state.

These synthetic results set the condition for interpreting the real-cohort experiments. Geometry alone is insufficient justification for a graph architecture; the stronger justification is local tumor-neighborhood response information that a nodewise forecaster misses. The real-data experiments below therefore evaluate both endpoint calibration and whether better edge definitions make message passing useful in the I-SPY2/ACRIN graph-bearing cohort.

B. ENDPOINT CALIBRATION CORRECTED THE REAL-COHORT BASELINE FAILURE MODE

The scheduled-sampling baseline had a specific endpoint failure rather than a global collapse of the graph rollout. It preserved the spatial rollout representation, but its deterministic center substantially underpredicted final tumor burden. In the held-out $T_0 \to T_3$ evaluation, observed mean FTV was 24.94 mL while the baseline deterministic mean was 12.43 mL, giving a mean bias of -12.51 mL and mean absolute error (MAE) of 12.88 mL. This is the central diagnostic result: a graph trajectory can remain geometrically plausible while still being miscentered in total active tumor volume.

Adding endpoint-level FTV and active-burden objectives corrected that failure mode while keeping the graph rollout backbone fixed. The first endpoint-calibrated model, Endpoint+Active, predicted a deterministic T_3 mean of 23.21 mL, reducing bias to -1.73 mL and MAE to 5.88 mL. The paired MAE reduction was 7.00 mL (95% bootstrap CI: 6.07–7.95), a 54.3% reduction in deterministic FTV error. This model's $T_0 \to T_3$ deterministic bias was near zero relative to the baseline and had a bootstrap mean CI of -2.94 to -0.59 mL.

The FTV correction preserved tumor-cloud geometry. SWD changed from 1.291 mm to 1.294 mm, Chamfer distance from 2.837 mm to 2.843 mm, Dice from 0.1338 to 0.1331, and surface Dice from 0.4631 to 0.4622. The large change was instead in active tumor burden: active-node-count absolute error decreased from 58.00 to 1.70 supervoxels. Figure 6 shows the same pattern at the patient level: endpoint calibration moves predictions toward the identity line while preserving the expected residual spread among high-volume tumors. The low absolute Dice values reflect this sparse transported-supervoxel cloud evaluation; the model forecasts active transported supervoxels rather than dense segmentation masks, so Dice is interpreted as a conservative overlap diagnostic rather than as a segmentation-quality endpoint. This is the first empirical signature of endpoint-calibrated spatial forecasting: the clinically queried endpoint changed substantially, while the spatial state metrics were essentially preserved. Table 5 reports the deterministic endpoint and spatial metrics for this failure-mode correction.

C. THE SAME MONTE CARLO SAMPLER BECAME SHARPER AND BETTER CALIBRATED

The decisive test was whether the uncertainty distribution improved when the sampler was held fixed and only the deterministic rollout center was changed. Under the same residual buckets, target-patient exclusion, active-mask sampling, and conformal interval calculation, the $T_0 \to T_3$ MC-mean FTV MAE decreased from 15.26 mL to 7.61 mL. The paired MAE reduction was 7.65 mL (95% CI: 7.11–8.20). This shows that

TABLE 4. **Controlled synthetic $T_0 \to T_3$ tests of when message passing helps. Deltas are Hybrid-Edge $k = 8$ minus the no-edge endpoint model; negative values are better for MC MAE, CRPS, interval width, and Chamfer.**

Synthetic cohort	Intended graph signal	No-edge MC MAE (mL)	Hybrid-Edge $k = 8$ MC MAE (mL)	Δ MC MAE (mL)	Δ CRPS	Δ width (mL)	Δ MC Chamfer (mm)
Node-local	Future response mostly readable from node-local features; message passing is expected to add little.	12.79	12.94	+0.15	-0.04	+0.99	-0.47
Neighbor-coupled	Future response partly depends on local neighbor active-node state, resistance, and burden.	24.48	24.88	+0.40	-0.20	-1.02	-0.62
Latent spatial-field	Hidden smooth regional response field is observed only through noisy node-local proxies.	35.29	29.86	-5.43	-3.53	-24.14	-0.41

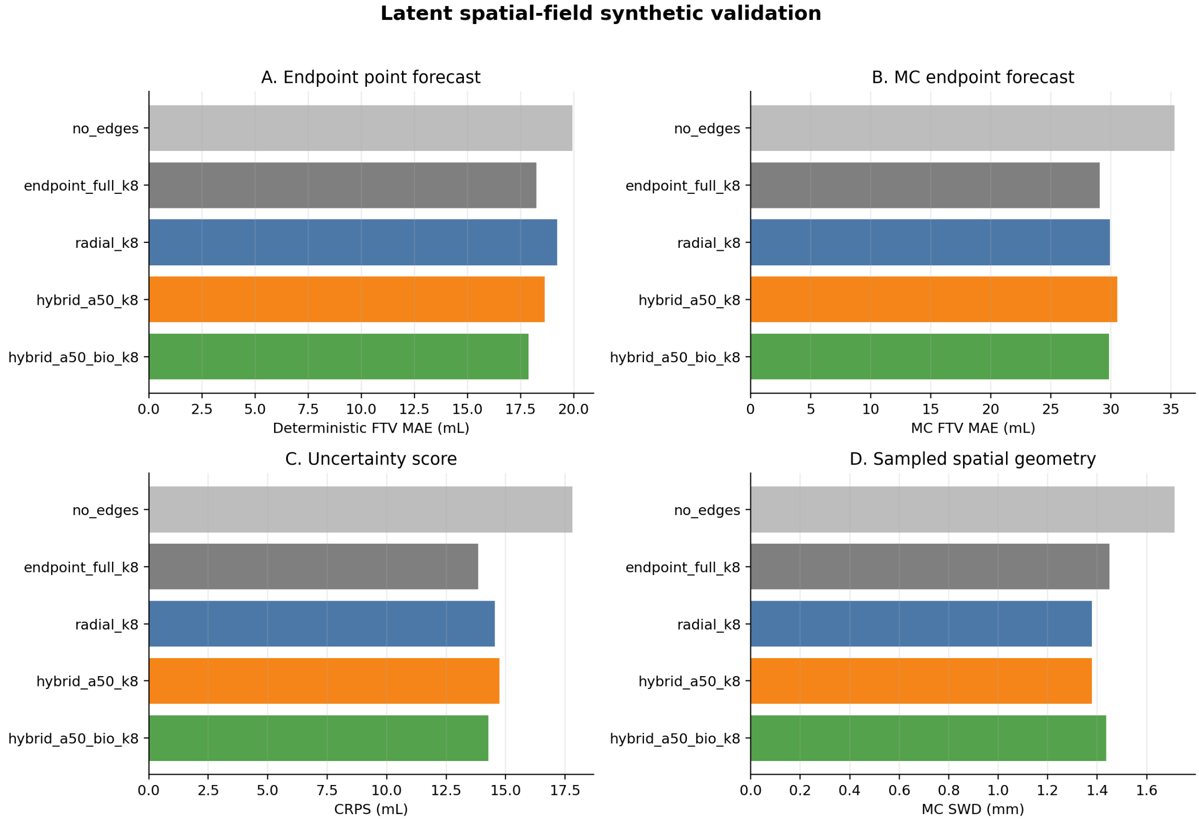

FIGURE 5. **Synthetic latent spatial-field validation. The no-edge control is competitive when response is node-local, but it degrades when future response depends on a hidden spatial field. In that regime, Hybrid-Edge $k = 8$ uses local neighborhood information to improve MC endpoint accuracy, CRPS, interval sharpness, and sampled-cloud geometry.**

the baseline MC layer was being forced to compensate for a biased center.

The interval behavior gives the strongest evidence that uncertainty calibration improved. Raw 90% coverage increased from 0.766 to 0.876 before conformal expansion (paired increase: 0.109; 95% CI: 0.075–0.145). At the same time, raw interval width decreased from 56.87 mL to 24.42 mL, and conformal interval width decreased from 59.51 mL to 27.95 mL. The paired conformal-width reduction was 31.57 mL (95% CI: 30.41–32.80). This combination matters because coverage and sharpness improved together. The resulting distribution was both better centered and narrower.

The full probabilistic score moved in the same direction. CRPS decreased from 8.35 to 5.16, indicating that the retained distribution placed more probability mass near the observed FTV. MC active-node-count error decreased from 40.78 to 2.69 supervoxels, aligning the sampled active graph burden with the volume calibration result. Table 6 reports the corresponding Monte Carlo distribution metrics.

The final retained graph-family model also improved calibration beyond the single 90% interval summary. In $T_0 \to T_3$ prediction, empirical coverage rose from 0.340 to 0.482 at nominal 50% intervals, from 0.766 to 0.877 at nominal 90% intervals, and from 0.875 to 0.925 at nominal 95% intervals. Calibration remains imperfect, but the final model moves the whole central-interval family toward nominal coverage rather

TABLE 5. $T_0 \rightarrow T_3$ **held-out deterministic endpoint results. Endpoint calibration removes the original low-FTV bias while preserving the main sampled-cloud geometry metrics.**

Metric	Baseline	Calibrated
Predicted FTV mean (mL)	12.43	23.21
Observed FTV mean (mL)	24.94	24.94
FTV bias (mL)	-12.51	-1.73
FTV MAE (mL)	12.88	5.88
Active-node-count abs. error	58.00	1.70
SWD (mm)	1.291	1.294
Chamfer (mm)	2.837	2.843
Dice	0.1338	0.1331

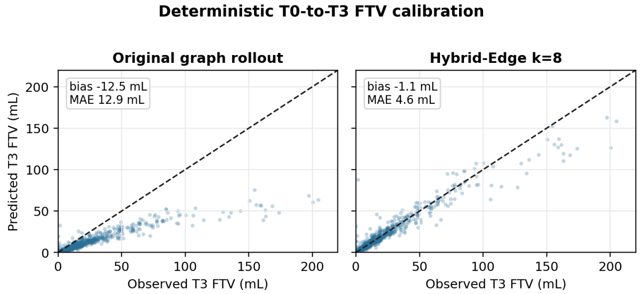

FIGURE 6. $T_0 \rightarrow T_3$ **deterministic FTV predictions versus observed** T_3 **FTV. The final retained graph model removes the large low-bias center seen in the scheduled-sampling baseline.**

TABLE 6. $T_0 \rightarrow T_3$ **held-out conditional Monte Carlo results using the same residual sampler. Endpoint calibration improves MC-mean FTV error, coverage, interval sharpness, CRPS, and active-node burden.**

Metric	Baseline	Endpoint-calibrated
MC-mean FTV MAE (mL)	15.26	7.61
MC-median FTV MAE (mL)	10.69	6.08
Raw 90% coverage	0.766	0.876
Raw 90% width (mL)	56.87	24.42
Conformal 90% width (mL)	59.51	27.95
CRPS	8.35	5.16
MC active-node-count abs. error	40.78	2.69
SWD (mm)	1.320	1.207
Chamfer (mm)	3.656	3.485
Dice	0.047	0.051

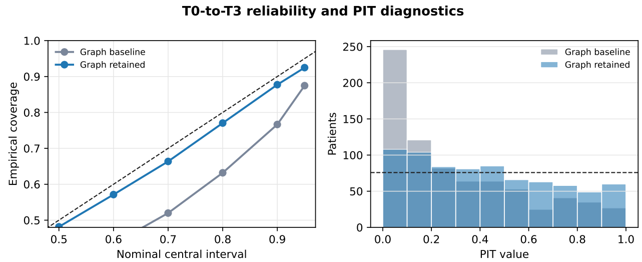

FIGURE 7. $T_0 \rightarrow T_3$ **reliability and PIT diagnostics for the baseline and final retained graph MC distributions. The final retained model improves central-interval coverage across nominal levels and makes the PIT distribution less boundary-concentrated.**

than only correcting one chosen interval. Figure 7 shows the coverage curve and the corresponding probability integral transform (PIT) diagnostic for the primary $T_0 \rightarrow T_3$ horizon.

D. THE ROLLOUT PRODUCED GRAPH-NATIVE STATE OUTPUTS

Only the graph rollout returns sampled active supervoxel clouds, so it can be evaluated with active-node and spatial cloud metrics in addition to FTV. Across final-visit conditioning buckets, endpoint/active-burden calibration reduced MC active-node count error by more than 93% while preserving or slightly improving sampled-cloud geometry (Table 7). This is the graph-native result that separates the model from a scalar FTV correction: endpoint calibration improves both the FTV center and the sampled active tumor-state center.

Table 8 then evaluates the calibration hierarchy that led to the endpoint-calibrated graph-family configuration.

Figure 8 visualizes the same endpoint-calibration mechanism by comparing deterministic bias, raw coverage, and interval width.

E. EDGE-AWARE NEIGHBORHOODS SELECTED THE FINAL GRAPH MODEL

The biology-calibrated model established that the rollout needed endpoint and active-burden supervision. The next question was whether message passing itself could be made more useful by changing the graph neighborhoods and edge attributes. This ablation compared no-edge, spatial, radial, feature-based, hybrid spatial-feature, and radial imaging-feature edge variants under the same held-out fold protocol. The final retained model for the paper is therefore Hybrid-Edge $k = 8$, a hybrid spatial-feature neighborhood model with radial imaging-feature edge attributes, rather than the earlier endpoint/active-calibrated graph.

The full edge-family search is kept in the Supplementary Material so the main paper can focus on the three reviewer-facing landmarks: the original graph rollout, the endpoint-calibrated graph, and the final edge-aware graph. In the full ablation, the radial imaging-feature $k = 8$ variant had the lowest scalar MC-mean FTV MAE by a small margin (5.52 mL versus 5.56 mL), but Hybrid-Edge $k = 8$ gave the strongest overall publication balance: nearly identical endpoint accuracy, the narrowest conformal intervals, the best raw 90% coverage among the top endpoint models, and preserved sampled-cloud geometry. Compared with the no-edge endpoint model, Hybrid-Edge $k = 8$ lowered deterministic FTV MAE by 0.65 mL, MC-mean FTV MAE by 1.02 mL, CRPS by 0.56, conformal width by 4.12 mL, and MC Chamfer by 0.043 mm. This is the clearest evidence that edges carry measurable signal: when the neighborhood is constrained to local spatial and feature-relevant neighbors, message passing improves the endpoint uncertainty layer without damaging graph-state geometry.

Figure 9 shows the same model-selection result as an endpoint–geometry tradeoff. This view makes the selection rule explicit: Hybrid-Edge $k = 8$ was retained because it

TABLE 7. Graph-native final-visit Monte Carlo state outputs. Active-node error is MC mean absolute error in sampled active supervoxel count. Spatial deltas are endpoint-calibrated model minus baseline; negative is better for SWD and Chamfer, positive is better for Dice. Scalar FTV baselines have no active-node or sampled-cloud output.

Bucket	Base active err.	Endpoint active err.	SWD Δ (mm)	Chamfer Δ (mm)	Dice Δ
$T_0 \to T_3$	40.78	2.69	-0.113	-0.171	+0.004
$T_1 \to T_3$	41.31	2.65	-0.108	-0.178	+0.003
$T_2 \to T_3$	41.64	2.66	-0.085	-0.151	+0.003

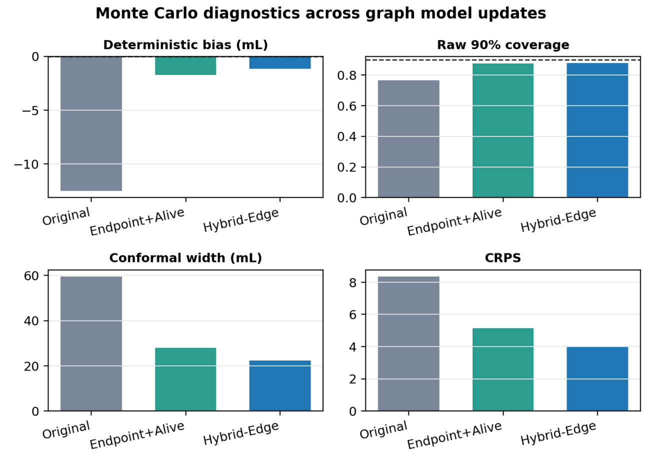

FIGURE 8. Monte Carlo diagnostics across graph model updates. Endpoint/active-burden calibration corrects the original low-bias center, and the final edge-aware model further sharpens CRPS and conformal interval width.

stayed near the best scalar endpoint error while giving the strongest balance of interval sharpness and sampled-cloud geometry.

F. TASK-ADAPTED SCALAR AND HYBRID BASELINES

Task-adapted FTV controls are necessary because a future-volume forecasting paper needs scalar as well as graph-family comparisons. Table 10 therefore separates simple scalar centers, scalar residual Monte Carlo, the two graph rollout centers, and the hybrid graph-plus-scalar readout. The deterministic carry-forward and trend baselines use only the available FTV history. For $T_0 \to T_3$, the linear and log-linear trends reduce to carry-forward because only the baseline visit is observed; at later starting visits they use the observed FTV history through the conditioning visit.

The scalar controls are intentionally strong. A last-observed FTV center with out-of-fold residual Monte Carlo achieved 4.82 mL MC-mean MAE, 0.898 raw coverage, 18.60 mL raw width, and 3.76 CRPS for $T_0 \to T_3$. A stronger regularized temporal scalar center, fit out-of-fold from observed FTV history, subtype, collection, and start visit, achieved 4.70 mL MC-mean MAE, 0.891 raw coverage, 18.98 mL raw width, and 3.84 CRPS. A nonlinear kernel version was less stable in this sample (15.82 mL MC-mean MAE), so it is retained only as a stress-test scalar control in Figure 10. These scalar FTV metrics are better than the graph rollout alone,

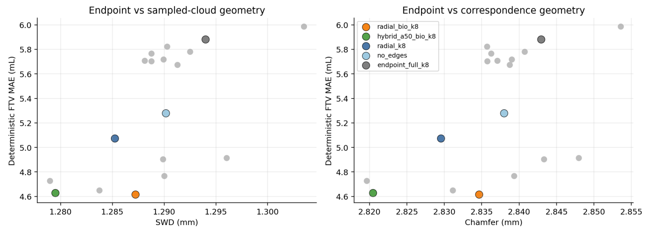

FIGURE 9. Endpoint-geometry tradeoff for the latest edge-aware graph-family ablation. The final model is selected for cross-layer balance rather than only for the lowest scalar FTV error.

even after the edge-aware update. The graph contribution is therefore structured-state forecasting rather than scalar burden dominance. The graph model contributes the structured spatial tumor-state forecast and active-node dynamics, while the scalar residual layer provides a direct patient-level FTV calibration. Therefore, the graph model is not positioned as the best scalar FTV regressor; its added value is the structured tumor-state forecast, with scalar or hybrid calibration available for scalar-only endpoint use.

The hybrid readout combines these roles. An out-of-fold ridge center using the graph prediction, last-observed FTV, and T_0 FTV improved $T_0 \to T_3$ MC-mean MAE to 4.50 mL and slightly narrowed raw interval width to 18.20 mL. Across $T_0 \to T_3$, $T_1 \to T_3$, and $T_2 \to T_3$, hybrid MC-mean MAE was 4.50, 4.41, and 4.23 mL, respectively (Figure 10). This gives the cleanest final modeling strategy: retain the graph rollout for patient-specific spatial evolution, and use scalar or hybrid residual calibration when the endpoint is a scalar FTV distribution.

G. IMPROVEMENT PERSISTED ACROSS CONDITIONING TIMES, SOURCES, AND PATIENTS

The robustness check used the same held-out full-cohort MC rollout outputs with shared residual pools and shared conformal calibration across subgroups. Across the three final-visit horizons, Hybrid-Edge $k = 8$ raw 90% coverage was 0.860–0.877, conformal coverage was 0.901, and conformal widths were 22.38–25.61 mL. Paired MC-mean FTV MAE reductions versus the original graph rollout were 8.13–9.70 mL across $T_0 \to T_3$, $T_1 \to T_3$, and $T_2 \to T_3$ conditioning, showing that the uncertainty gain persisted beyond the baseline-to-final rollout.

Source-stratified robustness was evaluated without refitting source-specific models. The same retained full-cohort model and residual MC protocol were scored separately on ACRIN-6698 and I-SPY2 patients. This is not independent external validation, but it is a harder internal check than reporting only the pooled cohort. Table 11 shows stable interval behavior across the two sources: ACRIN-6698 had lower MC-mean FTV error, while I-SPY2 was harder by scalar error but had nearly identical raw coverage and conformal width.

A small independent-cohort stress test was then run on 11 graph-ready Breast-MRI-NACT-Pilot patients using the same source-trained paper-family graph checkpoints and source-

TABLE 8. $T_0 \rightarrow T_3$ **endpoint-calibration hierarchy. The baseline is the geometry-first graph rollout; FTV-only calibration corrects the scalar endpoint; FTV-plus-active calibration also constrains active-node burden. The high-volume tail is the top decile of observed** T_3 **FTV (≥58.79 mL; 76 patients). SV denotes supervoxels.**

Model	Det. bias (mL)	Det. MAE (mL)	MC MAE (mL)	Raw cov.	Conf. width (mL)	CRPS	Active err. (SV)	Tail MC MAE (mL)
Baseline	-12.51	12.88	15.26	0.766	59.51	8.35	40.78	42.59
Endpoint+Active	-1.73	5.88	7.61	0.876	27.95	5.16	2.69	29.65
FTV only .010	-1.51	5.99	7.31	0.875	26.92	5.15	7.00	30.85
FTV .010 + active .002	-1.50	5.99	7.53	0.883	29.14	5.19	2.70	29.79

TABLE 9. $T_0 \rightarrow T_3$ **main graph-model progression. The large edge-family ablation is provided in the Supplementary Material; the main table keeps the three landmarks needed to interpret the paper claim. MC MAE is the absolute error of the Monte Carlo mean FTV.**

Model	Det. MAE (mL)	MC MAE (mL)	CRPS	Raw cov.	Conf. width (mL)	MC SWD (mm)	MC Chamfer (mm)
Original graph rollout	12.88	15.26	8.35	0.766	59.51	1.320	3.656
Endpoint+Active	5.88	7.61	5.16	0.876	27.95	1.207	3.485
Hybrid-Edge $k = 8$	**4.63**	**5.56**	**4.00**	**0.877**	**22.38**	**1.172**	**3.449**

TABLE 10. **Task-adapted** $T_0 \rightarrow T_3$ **FTV baselines. The scalar trend rows are deterministic point centers. MC rows use out-of-fold residual sampling. Linear and log-linear trend centers equal carry-forward for this bucket because only** T_0 **is observed.**

Model	Role	Det. MAE (mL)	MC MAE (mL)	Raw cov.	Raw width (mL)	CRPS
Last-observed FTV	scalar carry-forward center	4.81	—	—	—	—
Linear FTV trend	scalar trend center	4.81	—	—	—	—
Log-linear FTV trend	scalar trend center	4.81	—	—	—	—
Last-observed scalar MC	scalar center with residual MC	4.81	4.82	0.898	18.60	3.76
Ridge temporal scalar MC	regularized temporal scalar center with residual MC	4.66	4.70	0.891	18.98	3.84
Log-ridge temporal scalar MC	log-volume temporal scalar center with residual MC	4.82	4.89	0.896	20.11	4.05
Original graph rollout	spatial graph center with residual MC	12.88	15.26	0.766	56.87	8.35
Endpoint+Active	endpoint-calibrated graph center with residual MC	5.88	7.61	0.876	24.42	5.16
Hybrid-Edge $k = 8$	final edge-aware graph center with residual MC	4.63	5.56	0.877	19.21	4.00
Hybrid graph+scalar MC	ridge center using graph, last-observed, and T_0 FTV	4.48	4.50	0.897	18.20	3.71

TABLE 11. **Source-stratified** $T_0 \rightarrow T_3$ **robustness for the retained Hybrid-Edge** $k = 8$ **MC rollout. The same full-cohort model and residual calibration protocol are used; this is source-stratified internal robustness, not independent external validation.**

Source	n	MC MAE (mL)	CRPS	Raw cov.	Conf. width (mL)
ACRIN-6698	202	4.55	2.93	0.881	21.78
I-SPY2	556	5.93	4.39	0.876	22.59

cohort residual MC calibration. This analysis was designed to test whether the relative graph-family ordering persists under a collection shift, not to validate clinical generalization. Table 12 shows that Hybrid-Edge $k = 8$ was the strongest of the paper-family variants in this external test: it had the lowest deterministic FTV MAE, lowest MC-mean FTV MAE, and highest raw and conformal 90% coverage among the four retained graph-family comparators. The same table also shows the transfer limitation: all four endpoint-calibrated variants overpredicted T_3 FTV on this small cohort, so cohort-specific recalibration remains necessary before this result can be used as external validation for clinical deployment.

As a post hoc stratified evaluation of the same $T_0 \rightarrow T_3$ rollout, every molecular subtype showed a Hybrid-Edge $k = 8$ MC-mean FTV MAE reduction of at least 8.76 mL versus the original graph rollout. The HR-/HER2+ stratum was smallest

TABLE 12. **Independent Breast-MRI-NACT-Pilot** $T_0 \rightarrow T_3$ **stress test on 11 graph-ready external patients. All rows use source-trained paper-family graph checkpoints and source-cohort residual MC calibration; no NACT residuals are used for calibration. This is a preliminary external test, not powered definitive external validation.**

Model	Det. MAE (mL)	Det. bias (mL)	MC MAE (mL)	Raw cov.	Conf. cov.
Endpoint+Active	20.17	+19.71	23.51	0.455	0.455
No-edge endpoint	19.21	+18.82	22.11	0.455	0.455
Radial MRI-feature $k = 8$	16.70	+16.36	19.31	0.455	0.455
Hybrid-Edge $k = 8$	**12.40**	+12.13	**15.01**	**0.545**	**0.545**

and hardest ($n = 57$; Hybrid-Edge $k = 8$ MC-mean MAE 7.52 mL; raw coverage 0.912), so this evidence supports robustness of the average improvement while showing that endpoint difficulty varies across biologic strata. The subtype labels use hormone receptor (HR), HER2, and triple-negative breast cancer (TNBC) status. Burden-conditional calibration exposed the remaining failure mode. Below the top decile of observed T_3 FTV (< 58.73 mL), Hybrid-Edge $k = 8$ had 3.97 mL MC-mean MAE, 0.924 raw coverage, and 2.49 CRPS, compared with 12.22 mL, 0.771, and 4.94 for the original graph rollout. In the top 5% (≥ 84.88 mL; 38 patients), however, Hybrid-Edge $k = 8$ still had 31.57 mL MC-mean MAE and only 0.237 raw coverage. The high-volume tail is therefore improved but remains the main calibration limitation. Table 13 summarizes

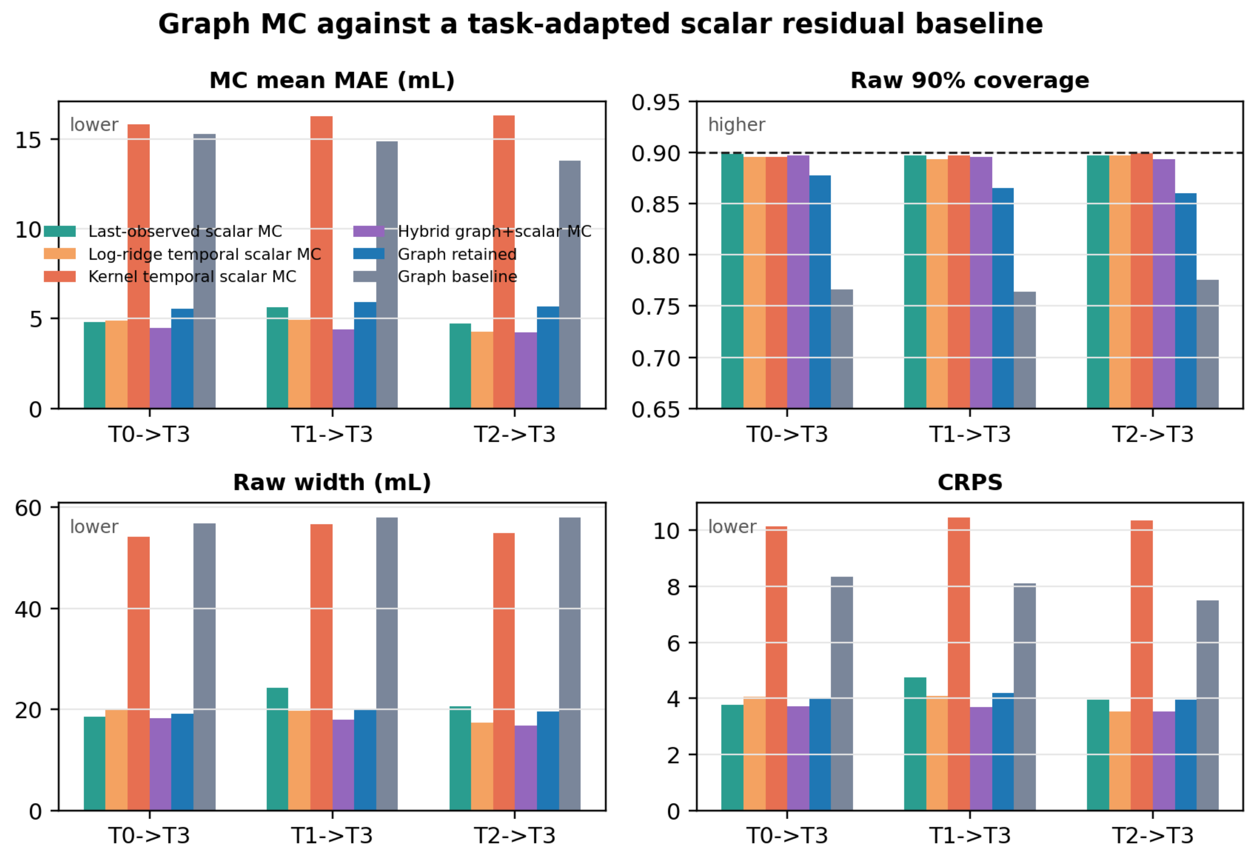

FIGURE 10. **Task-adapted scalar and hybrid controls for final-visit FTV uncertainty. Scalar temporal baselines use only observed scalar FTV history and metadata before applying out-of-fold residual Monte Carlo. The hybrid model combines the graph center, last-observed FTV, and T_0 FTV before applying the same scalar residual MC calibration.**

the horizon and subtype robustness checks, and Figure 11 shows the quartile-level coverage and CRPS behavior.

H. CLINICAL MRI-BURDEN MONITORING READOUTS

The retained Monte Carlo rollout also gives a clinically interpretable burden-monitoring readout. Before considering pathology, the direct imaging question is whether the future T_3 MRI is likely to show low or high residual enhancing tumor burden. Table 14 therefore evaluates two threshold probabilities from the MC samples: $P(T_3\ \mathrm{FTV} < 5\ \mathrm{mL})$ for low residual MRI burden and $P(T_3\ \mathrm{FTV} > 20\ \mathrm{mL})$ for substantial residual burden. The model readouts are compared with simple last-observed FTV rules because current FTV is itself a strong clinical baseline. The retained model identified low residual MRI burden with AUC 0.957 and high residual MRI burden with AUC 0.984. The last-observed FTV rules were similarly discriminative, which is an important boundary on the claim. The graph rollout's added value is the calibrated future-burden distribution and graph-native tumor-state output, while current FTV remains a strong threshold baseline. For the MC probability scores, the low-residual-burden readout had Brier score 0.061 with calibration intercept 0.396 and slope 1.334, while the high-residual-burden readout had Brier score 0.052 with calibration intercept -1.330 and slope 0.929. These calibration summaries support using the probabilities as imaging-burden scores, while still keeping them separate from pathology response probabilities.

The pCR analysis is kept secondary because this model forecasts MRI burden rather than pathology. Where pathology labels are available, the near-zero MRI-burden score had only a modest association with pCR (AUC 0.569), with observed pCR increasing from 0.28 to 0.40 across the lowest and highest score bins. This supports using the readout as an imaging-burden summary, with pathology response modeling reserved for a response-specific study. Figure 12 shows the corresponding decision-curve view for low residual MRI burden.

The same readout can be updated as additional MRI visits become available. Figure 13 summarizes the retained MC rollout when conditioning on T_0 only, T_0/T_1, and $T_0/T_1/T_2$. Low-burden discrimination remained high across serial conditioning (AUC 0.954–0.957), as did high-burden discrimination (AUC 0.982–0.985). Endpoint error and interval width were stable rather than monotonically narrower under the current residual-pool design, so the supported clinical use is serial uncertainty-aware MRI burden monitoring.

The two uncertainty visualizations in Figures 14 and 15 support complementary claims. The cohort-sorted bands show

TABLE 13. Calibration and stratified robustness checks from the final retained MC rollout, Hybrid-Edge $k = 8$. Horizon rows summarize final-visit conditioning buckets. Subtype rows are post hoc strata of the same held-out $T_0 \to T_3$ full-cohort rollout using the full-cohort MC calibration.

Check	Stratum	n	Final MC MAE (mL)	MAE reduction (mL; 95% CI)	Raw 90% cov.	Additional evidence
Horizon	$T_0 \to T_3$	758	5.56	9.70 (8.87–10.58)	0.877	Conformal coverage 0.901; conformal width 22.38 mL; CRPS 4.00.
Horizon	$T_1 \to T_3$	758	5.90	8.97 (8.23–9.76)	0.865	Conformal coverage 0.901; conformal width 25.61 mL; CRPS 4.18.
Horizon	$T_2 \to T_3$	758	5.68	8.13 (7.44–8.86)	0.860	Conformal coverage 0.901; conformal width 24.45 mL; CRPS 3.94.
Subtype	HR+/HER2+	128	4.90	8.76 (7.55–9.86)	0.906	Post hoc $T_0 \to T_3$ stratum; full-cohort residual calibration.
Subtype	HR+/HER2-	312	5.24	9.56 (8.35–10.97)	0.869	Post hoc $T_0 \to T_3$ stratum; full-cohort residual calibration.
Subtype	HR-/HER2+	57	7.52	10.82 (7.12–16.31)	0.912	Smallest and hardest subtype stratum by FTV error.
Subtype	TNBC	261	5.86	10.08 (8.68–11.68)	0.866	Post hoc $T_0 \to T_3$ stratum; full-cohort residual calibration.

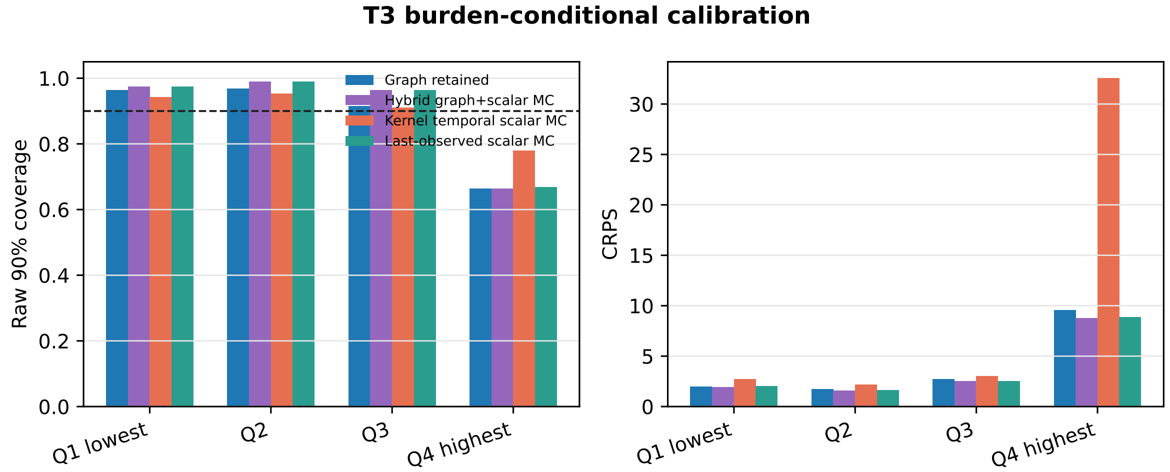

FIGURE 11. **Burden-conditional $T_0 \to T_3$ calibration by observed T_3 FTV quartile. Scalar and hybrid FTV readouts are strongest for scalar burden, while the retained graph distribution improves substantially over the original graph baseline but remains less calibrated in the high-volume tail.**

TABLE 14. Secondary clinically interpretable MRI-burden readouts for the $T_0 \to T_3$ cohort. The MC scores use the retained Hybrid-Edge $k = 8$ residual Monte Carlo samples; last-observed FTV is included as a clinical-use baseline, and these endpoints are not pathology response claims.

Readout	Endpoint	Score	AUC	Sens.	Spec.	PPV	NPV
Low residual burden	T_3 FTV < 5 mL	$P(T_3 \text{FTV} < 5)$	0.957	0.755	0.970	0.870	0.937
Low residual burden	T_3 FTV < 5 mL	Last-observed FTV < 5 mL	0.961	0.679	0.980	0.900	0.920
High residual burden	T_3 FTV > 20 mL	$P(T_3 \text{FTV} > 20)$	0.984	0.967	0.914	0.844	0.983
High residual burden	T_3 FTV > 20 mL	Last-observed FTV > 20 mL	0.986	0.972	0.902	0.827	0.985

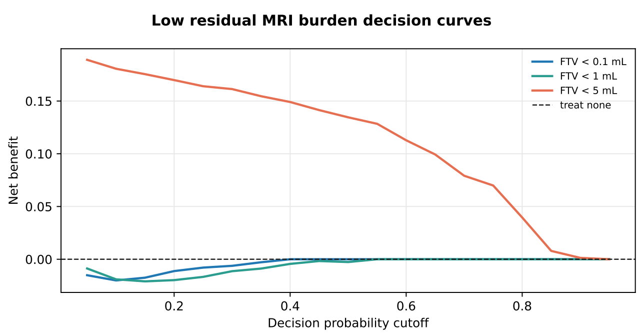

FIGURE 12. **Decision-curve analysis for retained-model low residual MRI burden scores. These curves are imaging-burden readouts from MC FTV samples rather than pathology response probabilities.**

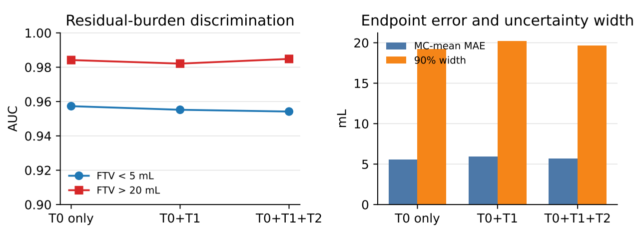

FIGURE 13. **Serial MRI-burden monitoring readouts for the retained model. Left: discrimination for low residual burden (T_3 FTV < 5 mL) and high residual burden (T_3 FTV > 20 mL) remains high as later visits are used for conditioning. Right: MC-mean FTV error and raw 90% interval width summarize the corresponding endpoint uncertainty.**

population-level behavior: across the held-out $T_0 \to T_3$ cohort, the final retained model shifts the MC mean toward the observed T_3 burden and compresses the predictive band, especially outside the lowest-volume cases. The patient trajectory examples then show the same mechanism at the individual level. Those panels span low, median, high, and large-improvement T_3 cases, so they serve as a diagnostic of how the Monte Carlo distribution changes over visits rather than a separate performance estimate. Together, the figures show that the retained model provides a patient-specific distribution over the future MRI tumor-burden trajectory rather than only a single final-visit point estimate.

VIII. DISCUSSION

This study supports a conditional view of graph message passing in longitudinal imaging. The synthetic experiments show that graph neighborhoods become useful under identifiable signal conditions. When synthetic response was node-local, the no-edge control remained competitive. When response depended on a hidden smooth spatial field, however, the final hybrid-edge graph model substantially improved MC endpoint error, CRPS, interval width, and sampled-cloud geometry. This is the key mechanistic result: message passing helps when local neighborhoods reveal response structure that isolated node features only partially observe.

The real-cohort results then show how this condition translates to breast DCE-MRI graph forecasting. Registration-consistent tumor graphs can be rolled forward through treatment, but the original rollout lacked a calibrated FTV uncertainty center. Its tumor clouds were plausible enough to evaluate, yet the FTV center was biased low and active mass was poorly calibrated. With the sampler, residual buckets, held-out fold protocol, and conformal calculation unchanged, endpoint-calibrated training produced sharper and better-centered distributions. The later edge-neighborhood ablation showed that the graph structure itself could also be improved: local hybrid spatial-feature neighborhoods with radial imaging-feature edge attributes gave the best balanced graph-family model.

For breast imaging, the clinical relevance is the ability to forecast an interpretable burden trajectory rather than only assign a response class. The most direct readouts are probabilities of clinically meaningful MRI-burden states, such as $P(T_3\mathrm{FTV} < 5\,\mathrm{mL})$ for low residual burden and $P(T_3\mathrm{FTV} > 20\,\mathrm{mL})$ for substantial residual burden. Expected FTV summarizes the likely future residual disease, while interval width and coverage communicate how much confidence should be placed in that forecast. These outputs could support routine response monitoring or trial-level assessment by flagging cases with likely low residual MRI burden, persistent high burden, or wide uncertain intervals. This positions the model as an uncertainty-aware burden-monitoring module: a step toward the kind of forecast that clinical or translational imaging workflows would need before clinical translation.

The scalar and hybrid controls define the boundary of the claim. For scalar FTV alone, a last-observed FTV center with out-of-fold residual Monte Carlo is a strong baseline, and the best scalar readout tested here comes from the hybrid center that combines the graph prediction with last-observed and T_0 FTV. This clarifies the role of the graph result. The graph rollout supplies a patient-specific spatial tumor-state forecast and active-node dynamics, while scalar residual calibration supplies an efficient burden correction for the final FTV distribution. The framework should therefore answer two linked questions: what future tumor state is expected, and what patient-level FTV distribution follows from the observed visits? The Hybrid-Edge $k = 8$ addresses the first while correcting the original burden failure; the hybrid residual readout gives the sharpest scalar answer to the second.

The calibration hierarchy provides the main theoretical evidence. Geometry alone produced plausible rollouts but failed as an FTV uncertainty center. FTV-only calibration nearly matched the best graph-family FTV MAE and CRPS, but left a larger active-node-count error. Adding active-burden calibration restored active-node consistency with nearly the same endpoint performance. The scalar and hybrid controls then show that the readout layer can be optimized separately from the graph state: a hybrid center is the best scalar FTV forecast, while the graph model remains the object that carries spatial tumor-state information.

Tables 8 and 9 clarify why Hybrid-Edge $k = 8$ is the retained graph configuration rather than simply the lowest-MAE ablation. The first calibration grid showed that FTV-only calibration could nearly match the endpoint error of Endpoint+Active, but left substantially worse active-node-count calibration. The later edge-neighborhood grid showed that radial imaging-feature edges had a slightly lower scalar MC-mean FTV MAE than Hybrid-Edge $k = 8$, but Hybrid-Edge $k = 8$ gave narrower conformal intervals, better raw coverage, and slightly stronger sampled-cloud geometry. That balance is essential because the method is intended to remain a graph rollout model with scalar FTV readout, instead of reducing the contribution to scalar FTV regression alone.

The remaining spatial results should be interpreted carefully. Endpoint calibration preserved SWD, Chamfer distance, and Dice, which is necessary for the graph-forecasting claim. However, the largest gains are clearly in tumor-burden calibration rather than in full spatial uncertainty. The current evidence therefore supports an endpoint-calibrated graph forecaster with preserved cloud geometry and a strong hybrid scalar uncertainty readout, while leaving full probabilistic three-dimensional tumor-shape forecasting as future work.

IX. CONDITIONS OF USE AND LIMITATIONS

Three limitations should be foregrounded before considering use outside the retrospective study setting. First, scalar FTV baselines are very strong: last-observed scalar residual MC reached 4.82 mL MC-mean MAE, and the hybrid graph-plus-scalar readout reached 4.50 mL, compared with 5.56 mL for the retained graph rollout alone. The graph model should therefore be interpreted as a structured tumor-state forecaster with a calibrated FTV readout, not as a replacement for task-adapted scalar calibration when the only endpoint is future FTV. Second, the highest-burden tail remains undercovered: in the top 5% of observed T_3 FTV, the retained graph had 31.57 mL MC-mean MAE and 0.237 raw 90% coverage. Third, the independent Breast-MRI-NACT-Pilot analysis contains only 11 graph-ready patients and is a stress test of transfer behavior,

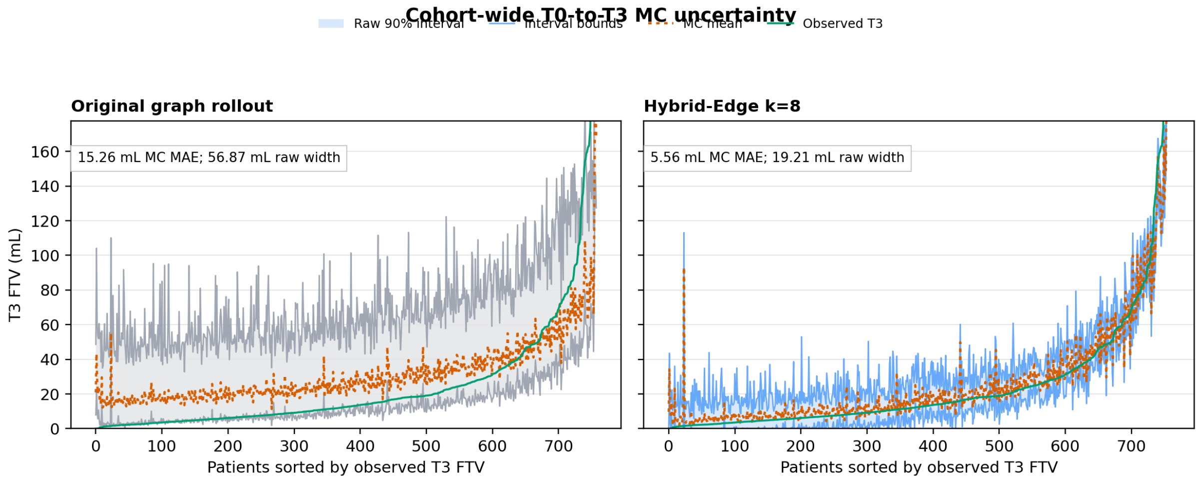

FIGURE 14. Cohort-wide $T_0 \to T_3$ MC uncertainty bands, with patients sorted by observed T_3 FTV. Side-by-side panels show the original graph rollout and the final retained model; shaded regions and same-color boundaries show raw 90% intervals, dotted orange lines show MC means, and solid green lines show observed T_3 FTV. The retained model produces a narrower and better-centered patient-level distribution while improving raw coverage and CRPS.

not powered external validation.

The method has clear conditions of use. It should work best when the graph preserves meaningful regional identity across visits, registration makes local neighborhoods interpretable, tumor response has spatial coherence, and node features only partially reveal future response. Under those conditions, message passing can denoise or propagate local response information, while endpoint calibration aligns the graph rollout with the clinically queried FTV endpoint.

It should help less when dynamics are purely node-local, edge neighborhoods are anatomically uninformative, segmentation or registration noise dominates local structure, or endpoint residuals are strongly non-exchangeable across burden or subtype strata.

The most important current technical failure modes are the high-volume tail and cross-cohort calibration shift. Even though the retained model improved average FTV error and calibration, it remains less reliable for very large residual tumor burdens. The independent 11-patient Breast-MRI-NACT-Pilot stress test also showed that Hybrid-Edge $k = 8$ transferred best among the paper-family graph variants, but still overpredicted endpoint FTV and achieved only 0.545 raw and conformal 90% coverage. This supports preliminary transfer of the relative model ordering, while identifying recalibration as an explicit target: future versions should test regularized FTV-stratified residual MC, subtype-aware calibration, tail-aware conformal correction, and external-cohort recalibration before using the system for high-burden or cross-cohort clinical readouts.

The supported interpretation is imaging-burden forecasting. The model predicts future MRI FTV and sampled tumor clouds under the available graph representation. Treatment counterfactuals, therapy-schedule optimization, and endpoints outside the longitudinal MRI sequence require separate modeling designs. The low residual imaging burden score reported above is therefore an exploratory association with pathology pCR where labels are available, and pathology response modeling would require a supervised response-specific calibration study. The modest pCR association reinforces the boundary of the present claim: this paper evaluates imaging-burden forecasting rather than clinical response classification. The synthetic cohorts validate mechanisms under controlled assumptions, and external clinical validation remains the next step before clinical translation. Before deployment, the forecast would also need prospective validation of whether uncertainty-aware FTV trajectories change reader confidence or trial-level response assessment.

The residual Monte Carlo sampler is empirical and depends on the exchangeability of held-out residuals within conditioning buckets. The current MC samples also mix spatial uncertainty, active-node uncertainty, and scalar FTV residual uncertainty; a fuller uncertainty decomposition would estimate these components separately. The current real-data evaluation is primarily internal to the I-SPY2/ACRIN-derived cohort, and the Breast-MRI-NACT-Pilot analysis is a small preliminary external test rather than definitive external validation. Larger independent longitudinal breast MRI cohorts are required before clinical translation.

X. CONCLUSION

Controlled synthetic experiments showed that graph message passing is most useful when local neighborhoods carry hidden response information, and the real-cohort experiments showed that endpoint calibration converts a spatial graph rollout into a burden-calibrated tumor-state forecaster. The retained Hybrid-Edge $k = 8$ model reduced $T_0 \to T_3$ deterministic FTV MAE from 12.88 mL to 4.63 mL, reduced graph-based MC-mean FTV MAE from 15.26 mL to 5.56 mL, improved raw 90% coverage from 0.766 to 0.877, cut conformal interval width from 59.51 mL to 22.38 mL, and preserved sampled-cloud geometry. The scalar controls sharpen the final message: future FTV uncertainty is best treated as a calibrated burden readout on top of the spatial graph forecast, with scalar or hybrid centers

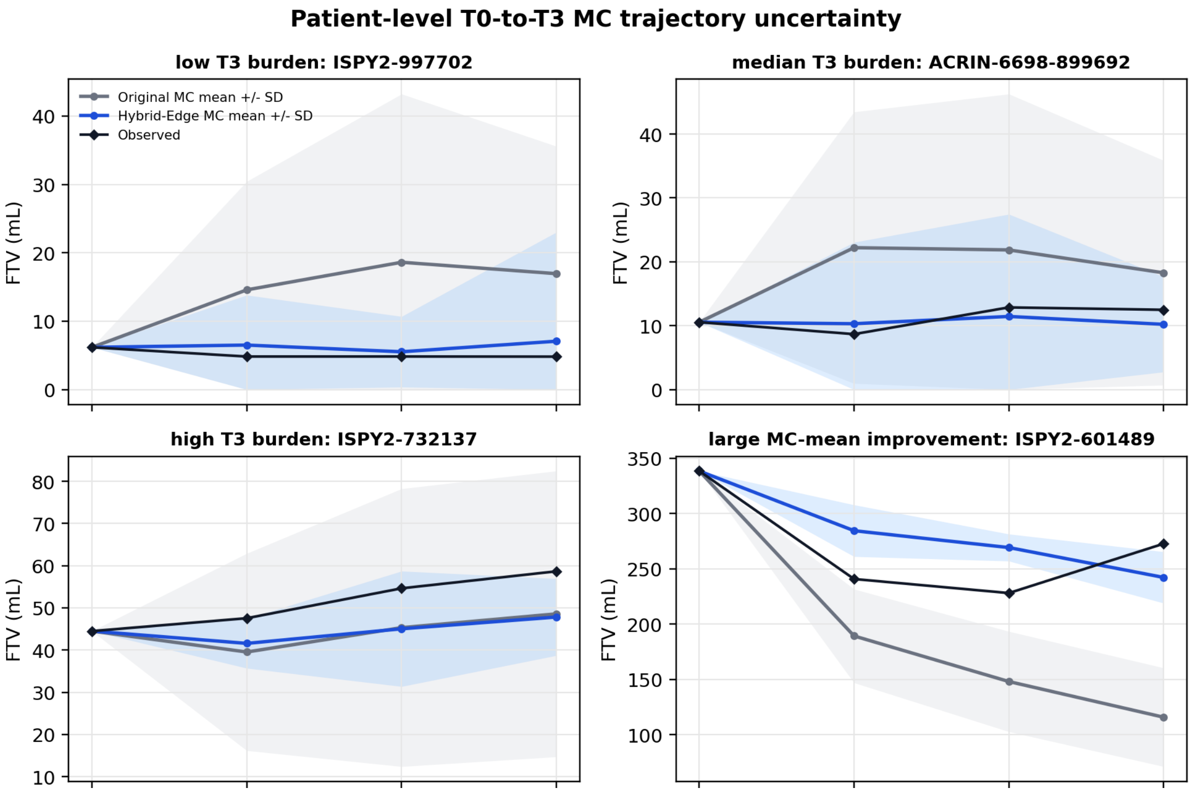

FIGURE 15. Representative $T_0 \rightarrow T_3$ Monte Carlo trajectory uncertainty examples. Gray shows the baseline MC mean and standard-deviation band, blue shows the final retained-model MC mean and standard-deviation band, and black diamonds show observed FTV.

remaining useful when only scalar FTV is needed. The graph-only model is not intended to replace scalar FTV baselines for scalar-only prediction; its contribution is a structured tumor-state forecast with calibrated endpoint readout. The threshold analyses likewise show that current FTV remains a strong clinical baseline; the added value of the proposed framework is the combination of calibrated future-burden probabilities with graph-native tumor-state forecasting. The central result is therefore methodological and practical: in longitudinal MRI graph rollouts, neighbor-aware spatial dynamics, endpoint calibration, and residual uncertainty calibration are separable layers that should be evaluated together.

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